

Rethinking case attraction on Ancient Greek infinitive clauses

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1 Introduction

Good morning. Nominal predicates and adjuncts modifying subjects of complement infinitive clauses in Ancient Greek (AG) usually agree in case with their domain local subject, which has morphological accusative case assigned in the *Accusativus cum Infinitivo* (*AcI*) construction. When the subject of the infinitive is not overtly expressed but the main clause has a dative or genitive object which shares the reference of the infinitive subject, these predicates or adjuncts may either occur in the accusative or “be attracted” to the case of the main clause’s object.

- (1) a. συμβουλεύει τῷ Ξενοφῶντι ἐλθόντα εἰς Δελφοὺς ἀνακοινῶσαι τῷ θεῷ περὶ τῆς πορείας.
He advises Xenophon to go to Delphi and ask the god about the travel. (Xen. *Anab.* 3.1.5)
- b. ἀφῆκε μοι ἐλθόντι πρὸς ὑμᾶς λέγειν τὰληθῆ.
He allowed me to go and speak the truth in front of you. (Xen. *Hell.* 6.1.13)

The literature on the subject, on the more philological side, from Port-Royal (Lancelot 1658, 1709) up to the Cambridge Grammar (Boas et al. 2019), have focused on listing the contexts in which attraction may occur, whereas the transformational and minimalist approaches of Lakoff (1970), Andrews (1971), Quicoli (1982), Spyropoulos (2005), Tantalou (2003), Sevdali (2007, 2013) have

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mainly occupied themselves with proposing the structure that would allow the phenomenon to occur. What factors influence the decision between either resolution in (1) remains, however, largely unknown.

[Self-deprecatingly] I myself have, in the last edition of this same event, tried to present the distributional patterns that make the attraction, (1-b), more likely, showing that there is strong correlation between attraction and the classes of the matrix and infinitive verbs and the relative distance and position of the controller and target of agreement.

Furthermore, I argued that these factors were *proxies* to unobserved values, namely semantic and pragmatic features of the discourse setting, and that even in the absence of seen contrasting factors, precisely the case in the pair above (1), the unseen discourse factors would be still causing one resolution to be at the very least more likely.

[Introductingly] I wasn't, however, quite convinced about neither the methods, the data, and, consequently, the quality of the results at which I had arrived. The main reason for this distrust is the fact that there could be interaction between the factors analysed which the methods employed were not well designed to account for, at least not with the amount of data available. To solve these shortcomings, we must take some steps back and build a model for what is the linguistic nature of case attraction

The aim of this paper is to walk through the assumptions present in my own and other's assessments of the phenomenon, as I believe most of them are in themselves right, build from these assumptions a causal system and show how much of this system is left unexplained with what we do know and, more importantly, I believe, what we **can** know and measure about ancient languages in general.

2 The data

Before stepping into the framework and analysis proper, a brief word on the empirical data supporting this research and any conclusion coming from it. The dataset is composed of 343 sentences drawn from the Attic literary *corpus* and Herodotus. The sentences were extracted from the Diorisis Corpus (Vatri and McGillivray 2018, 2020) with help of the project's querying tool, Diorisis Search (Vatri 2020-2022). Although comprehensive, the selection is by no means exhaustive.

Each sentence has been annotated for values of interest, including authorship, literary genre, dialect as well as class of matrix and infinitive verb, class of

controller and target, distance between constituents, etc. Parts of the annotation processes were conducted programmatically and later checked manually.

Every model used in this work was processed using R (R Core Team 2013), the packages tidyverse (Wickham et al. 2019), dagitty (Textor et al. 2016), ggdag (Barrett 2025), and rethinking (McElreath 2020), as well as the software Stan (Stan Development Team 2025a,b).

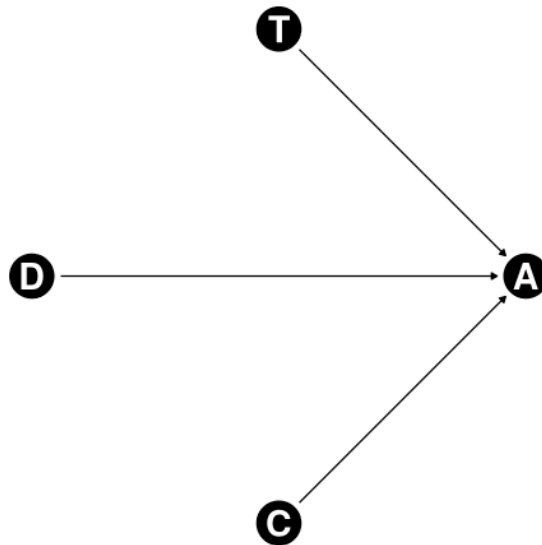
3 Nature of attraction

In the most abstract way possible, case attraction in infinitive clauses, exemplified by (1-b), is when two constituents, a controller and a target, produced in different domains of a sentence, one of the domains encompassing the other, covary in an specific feature, namely [*Case*]. This satisfies the definition of agreement in Corbett’s typological assessment of the phenomenon across languages (2006, pp. 4–7).

The lack of covariation of the [*Case*] feature, seen in (1-a), is generally understood as result of the application of **default** case concord rules of Ancient Greek. This assumes that the subject of infinitive clauses in Greek is always marked as accusative, even when not overtly expressed, and that convert constituents are also marked for [*Case*] and its case is also available for nominal concord.

The facts that the target has a choice of controller (either the overt oblique or de convert accusative) for agreement, that [*Case*] is a non-lexical features, that the agreement is optional, and that the domain is non-local favours the interpretation that the agreement type of case attraction is less canonical than the general agreement type of Ancient Greek, following the criteria for canonical agreement in the same work of Corbett (2006, pp. 8ff.).

If this assessment is sound, it is expected that attraction and non-attraction behave as competing agreement resolutions, the former being less canonical and thus more likely to be conditioned. Conditions concerning the semantic and pragmatic properties of the controller, its precedence in relation to the target and the class of predication (i.e.: class of the domain) have been identified by Corbett. Although properties of the target were not shown to be favouring any resolution in agreement mismatches in Corbett’s survey, case attraction might be one example of it, and I will hypothesize that properties of targets are competing causes of non-canonical resolutions. A model of agreement in this framework may be represented as a direct acyclic graph, in which Attraction ($\simeq$ Agreement) is caused by features of the **C**ontroller, **D**omain and **T**arget:



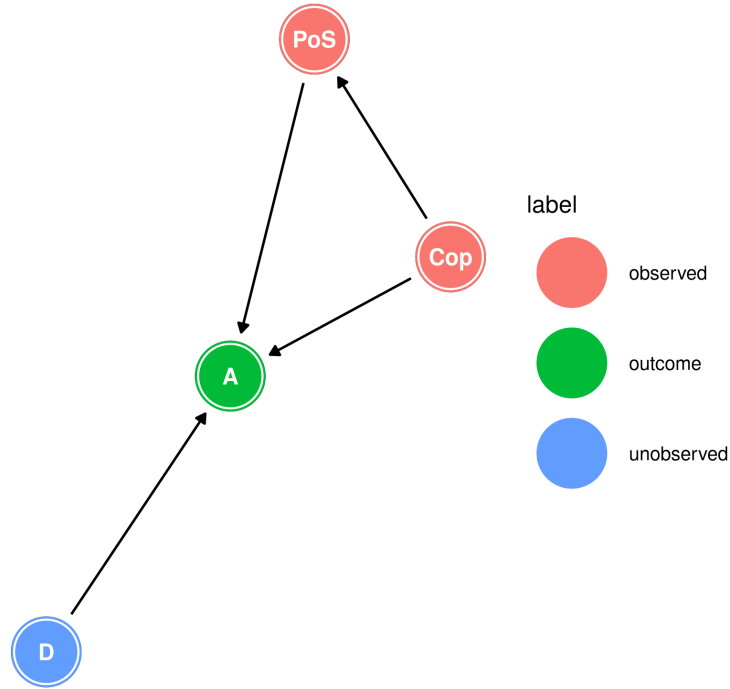
These Controller, Domain and Target nodes represent clusters of linguistic factors, some of which we may observe directly (part of speech of the target), some might be implied (animacy) and some may remain unobserved (discourse function, as Ancient Greek does not overtly mark focuses or topics in the morphosyntax, at least not dependably).

The three factors previously shown to correlate to case attraction represent each of these nodes:

- Copular infinitives imply a different domain of agreement, as they select different predication classes;
- Main verbs with semantic modality imply a different type of controller, as they have higher subjecthood;
- Word order and distance imply different status for the target, as targets more to the left of the sentence are both structurally closer to the controller and have a higher communicative function.

4 The domain of attraction

Firstly, Kühner and Gerth (1898) and others have associated case attraction with sentences in which the infinitival clause is a copular infinitive, *einai* or *genesthai*. Should the class of the infinitive favour either resolution of agreement? Probably not directly, as there is no reason linguistically it would. Copular infinitives, however, have a double effect on the agreement “causes”:



5 The controller of attraction

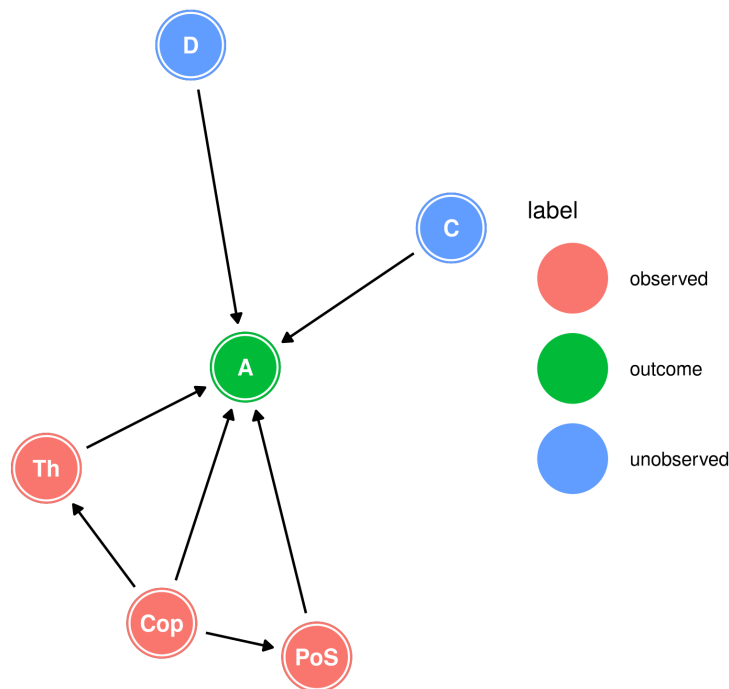
Another association is that between case attraction and impersonal verbs. I have, however, shown that this association does not exist with every single impersonal verb, and that the verb *δοκέω* almost never heads a sentence with case attraction. The impersonal verbs which are actually associated with case attraction are those carrying modal semantics, such as *ἐξαρκεῖ* and *ἐξεστί*. Is it likely that the nature of the matrix verb directly affects the choice of agreement? I don't think so, but it may indirectly do, by defining which features the controller of agreement, the oblique object, has. These verbs have very little semantic information and the theta role and grammatical function of their oblique object seems to depend on the infinitive verb. We may say that σοι in (1), for example, is the subject of the sentence (a quirk subject of sorts) and has the thematic role of agent, much in contrast if the sentence was constructed with a verb of order or advice like *συμβουλεύω*, to which the oblique represents a recipient of the action, and if the sentence is constructed by *δοκεῖ*, to which the oblique represents an experiencer, “the one who believes X to be better to do”.

- (1) ἄλλ' ἐξαρκεῖ σοι τὰ ἔργα αὐτῶν ὁρῶντι σέβεσθαι καὶ τιμᾶν τοὺς θεούς.

But it is enough for you to look at these deeds and respect the gods. (Xen. *Mem.* 4.3.13)

This, however, only holds if the infinitive verb is not copula, otherwise, the thematic role is also that of an experiencer, meaning that **Thematic Role** is a sister node of **Part of Speech**, as it is “caused’ by Copula.

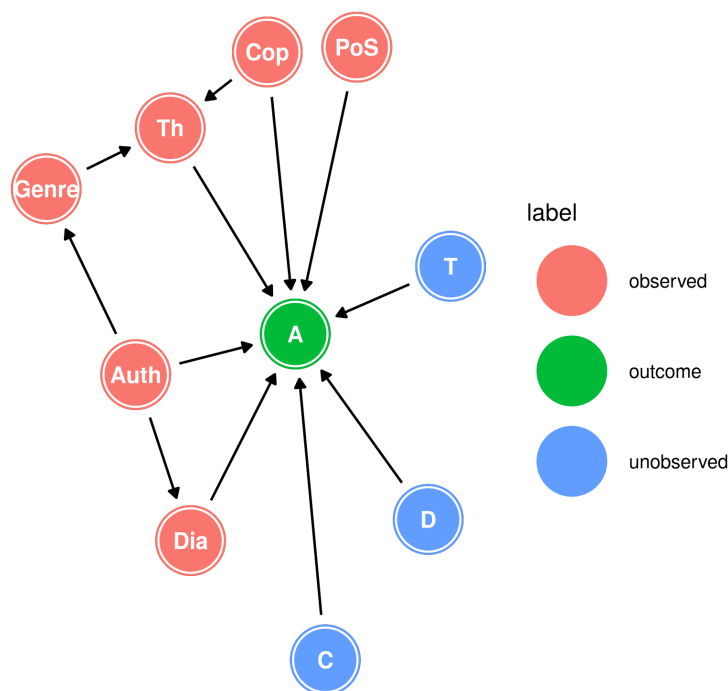
Adding to our previous DAG, the model will look like:



6 Intermezzo: Authors and dialects

I have argued previously that, in the corpus of Herodotus, Plato and Xenophon, the verbs of modal semantics were preferred by the authors when writing in the philosophical dialogue genre, and that this had implications if we were to estimate the effect of authorship or dialect on case attraction. If modality verbs affect the likelihood of attraction by means of selecting the thematic role of the controller, the author’s idiosyncrasies in the use of attraction or their dialect base likelihood of using attraction would be biased in the data by the effects of the selection of the main verb in sentences where attraction may appear.

We must thus add **Authorship**, **Genre** and **Dialect** to our causal model.



These last couple of additions have the following consequences:

- to estimate the direct effect of **Thematic** role, we must control for **Copula** and either by **Genre** or **Authorship**;
- to estimate the direct effect of **Copula**, we must control for **Part of Speech**, **Thematic** role, and either by **Genre** or **Authorship**;
- to estimate the direct effect of **Dialect**, we must control for **Authorship**;
- to estimate the direct effect of **Authorship**, we must control for **Dialect** and by either **Genre** or **Copula** and **Thematic** role.

7 The target of attraction

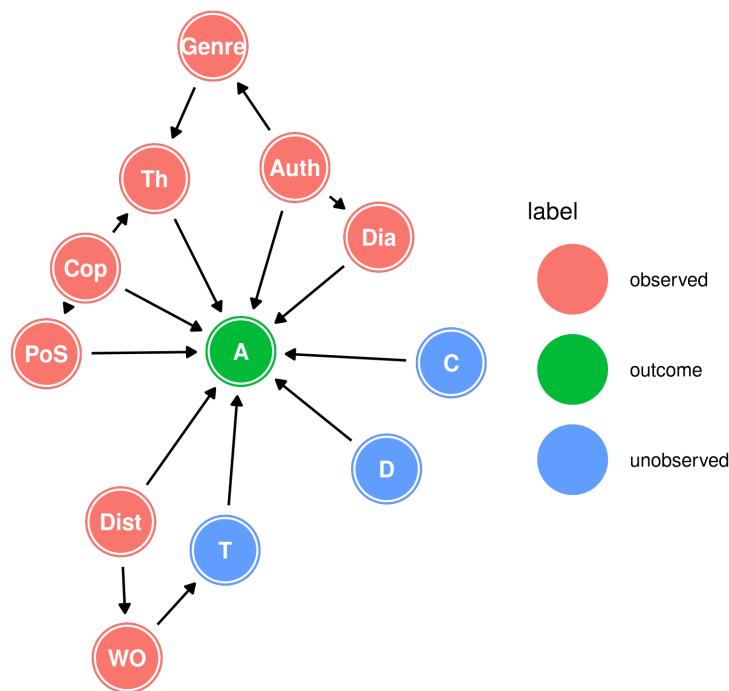
Finally, it has been long been proposed that case agreement is more likely when controller and target are very close to each other, and the likelihood diminishes as it grows. Furthermore, precedence of the target in relation to the controller seems to demand case attraction, such as in (1).

- (1) ὥς ἀγαθοῖς τε ὑμῖν προσήκει εἶναι.
That you may be good people. (Xen. Anab. 3.2.11)

Word order is highly associated in Ancient Greek with an important condition for non-canonical agreement: communicative function.¹ Although associated with, however, word order does not completely determines communicative function of the target: it is not enough to guarantee that a certain constituent is a topic or focus. In other words, Word Order would only affect case attraction indirectly, by ‘causing’ the communicative function of the target, function which is, however, unobservable. This means that the effect of word order over case attraction is statistically impossible to infer without a way to measure the communicative function of the target.

The effect of distance, on the other hand, is measurable, so long as we control for its effect on word order: distance changes word order, so we must block the non-causal path that flows from distance to attraction going through word order and communicative function.

In the model:



¹Allan (2012), Dik (1995, 2007), and Matic (2003)

8 Analysis

Fortunately, we don't need many different models to estimate the effects of many of the factors of interest. By applying the do-calculus, so long as we control for Word Order, we can estimate the direct effects of:

- Part of Speech of the target;
- Copular infinitive (i.e.: class of predication);
- Thematic role of the oblique object;
- Dialect;
- Authorship; and
- Distance

The following generalize linear model allows us to do so:

$$\begin{aligned}
A &\sim \text{Bernoulli}(p_i) \\
\text{logit}(p_i) &= \alpha + \beta_{\text{Cop}}\text{Cop}_i + \beta_{\text{Dist}}\text{Dist}_i + \beta_{\text{Wo}}\text{Prec}_i \\
&\quad + \beta_{\text{PoS}} \sum_{j=0}^{PoS_i-1} \delta_{j\text{PoS}} + \beta_{\text{Th}} \sum_{j=0}^{Th_i-1} \delta_{j\text{Th}} \\
&\quad + \beta_{\text{Auth}_i} + \beta_{\text{Dia}_i} \\
\alpha &\sim \text{Normal}(0, 1) \\
\beta_{\text{Cop}}, \beta_{\text{Dist}}, \beta_{\text{Prec}} &\sim \text{Normal}(0, 1) \\
\beta_{\text{PoS}}, \beta_{\text{Th}} &\sim \text{Normal}(0, 0.5) \\
\delta_{\text{PoS}}, \delta_{\text{Th}} &\sim \text{Dirichlet}(2) \\
\beta_{\text{Auth}_j} &\sim \text{Normal}(\bar{\beta}_{\text{Auth}}, \sigma_{\text{Auth}}) \\
\beta_{\text{Dia}_j} &\sim \text{Normal}(\bar{\beta}_{\text{Dia}}, \sigma_{\text{Dia}}) \\
\bar{\beta}_{\text{Auth}}, \bar{\beta}_{\text{Dia}} &\sim \text{Normal}(0, 1) \\
\sigma_{\text{Auth}}, \sigma_{\text{Dia}} &\sim \text{Exponential}(1)
\end{aligned}$$

To estimate the effect of *PoS* over Attraction, we can use the generalized linear model below. We can control for Author, as there is no reason to believe that any author from the corpus prefers copulas or any type of target. As it is usually agreed that typological hierarchies increase their effect monotonically by each of their nodes, but we do not know if their effect should be linear or not, a Dirichlet prior is suitable to express the effect of Part of Speech.

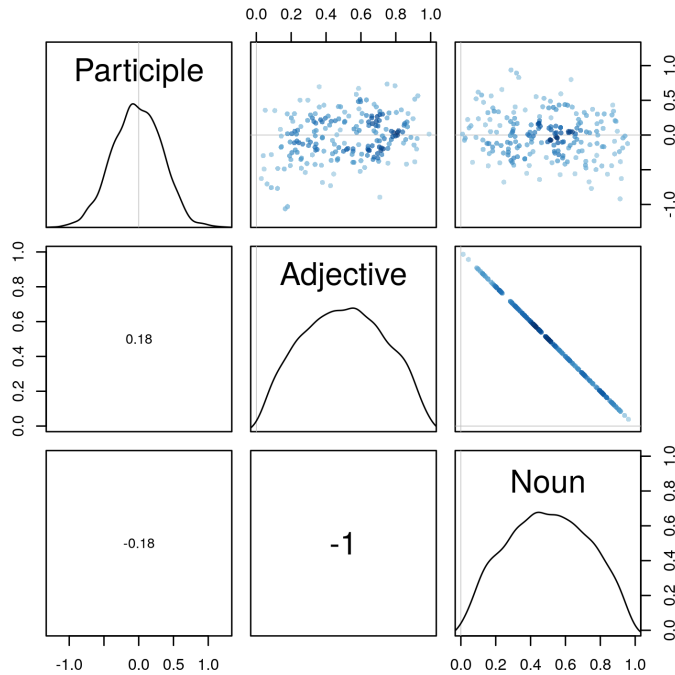
The priors are fairly flat, as there is no reason to believe we have weak or strong relationships in agreement phenomena. β_{PoS} is the general effect of the part of speech over the likelihood of Attraction, and δ is a vector with values

denoting how much going up from Participle to Adjective to Noun increases β_{PoS} . Participles do not change the effect, Adjectives add δ_1 and Nouns add $\delta_1 + \delta_2$, meaning that the effect is expected to be cumulative.

The estimated effects are described in the following table and plot.

Effect	mean	sd	5.5%	94.5%	rhat
β_{PoS}	-0.02	0.35	-0.58	0.53	1
δ_{Adj}	0.50	0.23	0.13	0.87	1
δ_{Noun}	0.50	0.23	0.13	0.87	1

PoS	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Participle	β_{PoS}	-0.02	0.35	-0.58	0.53	1
Adjective	$\beta_{\text{PoS}} + \delta_{\text{Adj}}$	0.48	0.45	-0.25	1.20	1
Noun	$\beta_{\text{PoS}} + \delta_{\text{Adj}} + \delta_{\text{Noun}}$	0.98	0.35	0.42	1.53	1

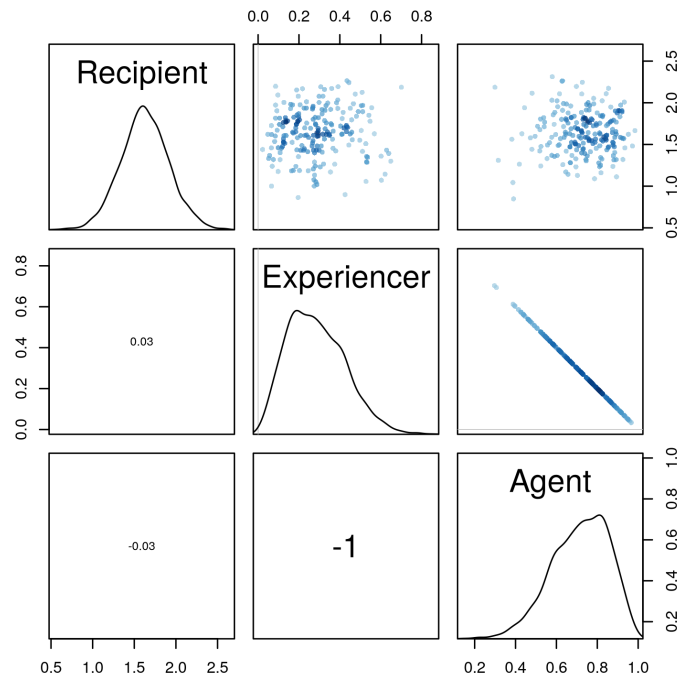


To put it into words, the data favours the idea that the Agreement and Predication Hierarchy holds for case attraction in Ancient Greek case attraction: as the target becomes more “nominal”, the likelihood of attraction increases

from basically neutral with participles up to highly positive. Notice that the model assumes that the effect of adjectives and nouns are negatively correlated, meaning that it does not discount the possibility that the bulk of the effect is in the passage from Participles to Adjectives, and going further up to a Noun makes little difference.

Effect	mean	sd	5.5%	94.5%	rhat
β_{Theta}	1.63	0.29	1.17	2.10	1
δ_{Exp}	0.28	0.14	0.08	0.53	1
δ_{Agent}	0.72	0.14	0.47	0.92	1

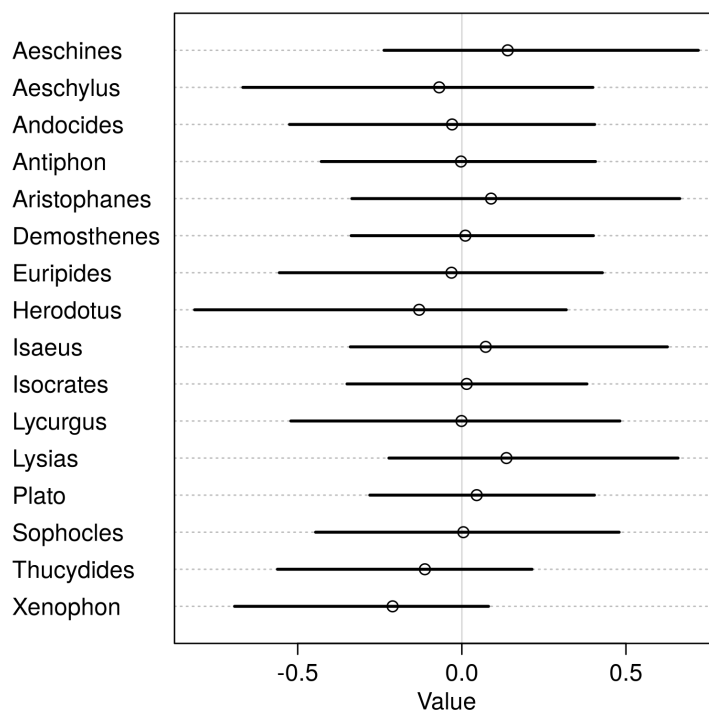
Theta	Cummulative effect	mean	sd	5.5%	94.5%	rhat
Recipient	β_{Rec}	1.63	0.29	1.17	2.10	1
Experiencer	$\beta_{\text{Rec}} + \delta_{\text{Exp}}$	1.91	0.32	1.40	2.44	1
Agent	$\beta_{\text{Rec}} + \delta_{\text{Exp}} + \delta_{\text{Agent}}$	2.63	0.29	2.17	3.10	1



Effect	mean	sd	5.5%	94.5%	rhat
β_{Copula}	1.58	0.37	0.98	2.19	1
β_{Dist}	-0.21	0.04	-0.27	-0.15	1

Effect	mean	sd	5.5%	94.5%	rhat
β_{Attic}	0.57	0.73	-0.56	1.74	1
β_{Jonic}	-1.20	0.84	-2.63	-0.01	1

Effect	mean	sd	5.5%	94.5%	rhat
β_{Xenophon}	-0.21	0.26	-0.69	0.08	1.00
$\beta_{\text{Herodotus}}$	-0.13	0.39	-0.81	0.32	1.00
$\beta_{\text{Thucydides}}$	-0.11	0.26	-0.56	0.21	1.00
$\beta_{\text{Aeschylus}}$	-0.07	0.34	-0.67	0.40	1.00
$\beta_{\text{Andocides}}$	-0.03	0.29	-0.52	0.40	1.00
$\beta_{\text{Euripides}}$	-0.03	0.33	-0.56	0.43	1.00
β_{Antiphon}	0.00	0.27	-0.43	0.41	1.01
β_{Lycurgus}	0.00	0.32	-0.52	0.48	1.00
$\beta_{\text{Sophocles}}$	0.00	0.31	-0.45	0.48	1.00
$\beta_{\text{Demosthenes}}$	0.01	0.23	-0.34	0.40	1.01
$\beta_{\text{Isocrates}}$	0.01	0.23	-0.35	0.38	1.00
β_{Plato}	0.05	0.22	-0.28	0.40	1.00
β_{Isaeus}	0.07	0.31	-0.34	0.63	1.00
$\beta_{\text{Aristophanes}}$	0.09	0.34	-0.34	0.66	1.00
$\beta_{\text{Aeschines}}$	0.14	0.31	-0.24	0.72	1.00
β_{Lysias}	0.14	0.29	-0.22	0.66	1.00



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